Chapter 15. The Spectral Theorem

Normal, symmetric and orthogonal operators

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Spectral theorems describe operators by orthonormal eigen-decompositions.

15.1 Notation and terminology

Motivation. One notation T^* covers transposes and adjoints; $T^* = T$, $T^*T = I$, $TT^* = T^*T$ define the special operators.

Notation 15.1.1

"$T \in \mathfrak{LM}$" means $T \in \mathfrak{L}(V, V)$ or $T \in \mathfrak{M}_{n,n}(F)$, with $\dim V = n$; a matrix T is regarded as L_T on $V = F^n$. We assume $V \neq 0$.

Notation 15.1.2

(a) For $F = \mathbb{C}$ and a non-degenerate Hermitian space (V, H), T^* means the adjoint $T^\star$; for a matrix A, $T^* = (L_A)^\star$. (b) For a non-degenerate quadratic space (V, B), T^* means the transpose T^t (usually written T^t).

Remark 15.1.3

(a) Dual maps do not appear again. (b) In $(\mathbb{C}^n, H)$, $[(L_A)^\star]_\mathcal{E} = \bar{J}^{-1}A^*\bar{J}$ ($J = [H]_\mathcal{E}$), which equals the adjoint matrix A^* when $\mathcal{E}$ is orthonormal (e.g. the Hermitian dot product). (c) Likewise $[(L_A)^t]_\mathcal{E} = J^{-1}A^tJ$ in (F^n, B), equal to A^t when $\mathcal{E}$ is orthonormal.

Observation 15.1.4

If $W \leq V$ is both T-invariant and T^*-invariant, then $(T|_W)^* = T^*|_W$.

Proof. With $\langle , \rangle = H$ or B: $\langle Tw, w' \rangle = \langle w, T^*|_W w' \rangle$ for $w, w' \in W$; apply Propositions 14.3.11, 13.9.3 on W (when $\langle , \rangle|_W$ is non-degenerate, e.g. inner products). ■

Definition 15.1.5 (Special operators)

For $T \in \mathfrak{LM}$: (a) if $T^* = T$, T is **self-adjoint** (**Hermitian**) in the case (V, H), and **symmetric** in the case (V, B); (b) if $T^*T = I$, T is **unitary** (case (V, H)) or **orthogonal** (case (V, B)); (c) if $TT^* = T^*T$, T is **normal**. ($A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is a **normal matrix** if $AA^* = A^*A$.)

Exercise 15.1.6

(a) Self-adjoint (symmetric) operators are normal.
(b) Unitary (orthogonal) operators are normal.

Solution. (a) $TT^* = T^2 = T^*T$.

(b) $T^*T = I$ forces T invertible with $T^{-1} = T^*$, so $TT^* = I = T^*T$. □

15.2 Normal operators

Motivation. On a Hermitian inner product space, orthonormally diagonalizable means normal.

Notation 15.2.1

Here $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional Hermitian inner product space.

Observation 15.2.2

If V has an orthonormal basis of eigenvectors of T, then T is normal.

Proof. $[T]_{\mathfrak{B}}^{\mathfrak{B}} = \mathrm{diag}(\lambda_i)$ and $[T^*]_{\mathfrak{B}}^{\mathfrak{B}} = \mathrm{diag}(\bar{\lambda}_i)$ (Proposition 14.3.12, $[H]_{\mathfrak{B}} = I$) commute. ■

Observation 15.2.3

T is normal iff $\langle Tv, Tw\rangle = \langle T^*v, T^*w\rangle$ for all v, w.

Proof. $\langle Tv, Tw\rangle = \langle v, T^*Tw\rangle$ and $\langle T^*v, T^*w\rangle = \langle v, TT^*w\rangle$; these agree for all v, w iff $T^*T = TT^*$. ■

Observation 15.2.4

If T is normal and $\lambda \in \mathbb{C}$, then $T - \lambda I$ is normal.

Proof. $(T - \lambda I)^* = T^* - \bar{\lambda} I$ (Observation 14.3.15, Exercise 14.3.16), and $(T - \lambda I)(T^* - \bar{\lambda} I) = TT^* - \bar{\lambda}T - \lambda T^* + |\lambda|^2 I = (T^* - \bar{\lambda} I)(T - \lambda I)$. ■

Observation 15.2.5

If T is normal and $Tv = \lambda v$, then $T^*v = \bar{\lambda} v$.

Proof. By Observations 15.2.3, 15.2.4: $0 = \langle (T - \lambda I)v, (T - \lambda I)v\rangle = \langle (T^* - \bar{\lambda} I)v, (T^* - \bar{\lambda} I)v\rangle$, so $(T^* - \bar{\lambda} I)v = 0$ by positive definiteness. ■

Theorem 15.2.6 (Spectral theorem, positive definite Hermitian case)

For $T \in \mathfrak{LM}$ (Notation 15.2.1) the following are equivalent: (1) T is normal; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1): Observation 15.2.2. (1)⇒(2): induct on n. Take a unit eigenvector $Tv = \lambda v$ and $W = \langle v\rangle^{\perp}$, $V = \langle v\rangle \oplus W$.

(i) W is T-invariant: for $u \in W$, $\langle v, Tu\rangle = \langle T^*v, u\rangle = \langle \bar{\lambda} v, u\rangle = \bar{\lambda}\langle v, u\rangle = 0$ (Observation 15.2.5).

(iii) W is T^*-invariant: $\langle v, T^*u\rangle = \langle Tv, u\rangle = \lambda\langle v, u\rangle = 0$.

(ii) By Observation 15.1.4, $(T|_W)^* = T^*|_W$, so $T|_W$ is normal. By induction W has an orthonormal eigenbasis of $T|_W$; add v. ■

Definition 15.2.7 (Maximal tori)

With $\mathbf{D}_n(\mathbb{C})$ the invertible diagonal matrices, $\mathbf{T}_{\mathbf{U}(n)} = \mathbf{U}(n) \cap \mathbf{D}_n(\mathbb{C})$ and $\mathbf{T}_{\mathbf{SU}(n)} = \mathbf{SU}(n) \cap \mathbf{D}_n(\mathbb{C}) = \mathbf{T}_{\mathbf{U}(n)} \cap \mathrm{SL}_n(\mathbb{C})$.

Exercise 15.2.8

(a) $\mathbf{T}_{\mathbf{U}(n)} = \{\mathrm{diag}(\lambda_1, \ldots, \lambda_n) : |\lambda_i| = 1\}$. (b) $\mathbf{T}_{\mathbf{U}(n)} \approx S^1 \times \cdots \times S^1 \approx \mathbf{U}(1) \times \cdots \times \mathbf{U}(1)$ (n factors). (c) $\mathbf{T}_{\mathbf{SU}(n)} \approx S^1 \times \cdots \times S^1$ ($n-1$ factors).

Solution. (a) A diagonal matrix has orthonormal columns iff $|\lambda_i| = 1$.

(b) $\mathrm{diag}(\lambda_i) \mapsto (\lambda_i)$.

(c) The last entry is determined by $\lambda_n = (\lambda_1 \cdots \lambda_{n-1})^{-1}$ (Exercise 11.9.13). □

Corollary 15.2.9 (Spectral theorem for $\mathbf{U}(n)$)

If $A \in \mathbf{U}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$ for some $U \in \mathbf{U}(n)$. Hence $\mathbf{U}(n) = \bigcup_{U \in \mathbf{U}(n)} U\, \mathbf{T}_{\mathbf{U}(n)}\, U^{-1}$.

Proof. $(L_A)^\star = L_{A^*}$ for the dot product (Remark 15.1.3(b)), so L_A is unitary, hence normal (Exercise 15.1.6), with an orthonormal eigenbasis $\mathfrak{B}$ (Theorem 15.2.6). Then $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} = U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$ (Remark 10.6.4, Proposition 14.1.12) with $U = [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{U}(n)$ (Exercise 10.6.9(5)). ■

Exercise 15.2.10

If $U \in \mathbf{U}(n)$, then $\alpha U \in \mathbf{SU}(n)$ for some $\alpha \in \mathbb{C}$.

Solution. $\det U = e^{i\theta}$; $\alpha = e^{-i\theta/n}$ gives $\det(\alpha U) = 1$, and $|\alpha| = 1$. □

Corollary 15.2.11 (Spectral theorem for $\mathbf{SU}(n)$)

If $A \in \mathbf{SU}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(n)}$ for some $U \in \mathbf{SU}(n)$; so $\mathbf{SU}(n) = \bigcup_{U \in \mathbf{SU}(n)} U\, \mathbf{T}_{\mathbf{SU}(n)}\, U^{-1}$.

Proof. Corollary 15.2.9 gives $U \in \mathbf{U}(n)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$, of determinant 1; replace U by $\alpha U \in \mathbf{SU}(n)$ (Exercise 15.2.10). ■

Exercise 15.2.12

In Corollary 15.2.9, can U always be chosen in $\mathbf{SU}(n)$, even when $A \notin \mathbf{SU}(n)$?

Solution. Yes: replace U by αU as in the proof of Corollary 15.2.11; the conjugate is unchanged. □

Exercise 15.2.13

If $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is normal ($AA^* = A^*A$), then $U^{-1}AU \in \mathbf{D}_n(\mathbb{C}) \cup \{\text{singular diagonal}\}$, i.e. $U^{-1}AU$ is diagonal, for some $U \in \mathbf{SU}(n)$.

Solution. Theorem 15.2.6 for L_A, then Corollary 15.2.9 and Exercise 15.2.12. □

Exercise 15.2.14

For $A = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right) \in \mathbf{SU}(2)$, find $U \in \mathbf{SU}(2)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(2)}$.

Solution. Eigenvalues $e^{\pm i\pi/4}$, unit eigenvectors $\frac{1}{\sqrt{2}}(1,1)^t$, $\frac{1}{\sqrt{2}}(-1,1)^t$; $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$, $\det U = 1$, $U^{-1}AU = \mathrm{diag}(e^{i\pi/4}, e^{-i\pi/4})$. □

15.3 Symmetric operators

Motivation. Real symmetric operators are orthonormally diagonalizable; this diagonalizes quadratic forms.

Notation 15.3.1

Through Exercise 15.3.11, $F = \mathbb{R}$, $(V, \langle\,,\rangle)$ is an n-dimensional real inner product space, and $T^t = T^*$.

Proposition 15.3.2

(a) A symmetric operator T has a real eigenvalue.
(b) A real symmetric matrix has a real eigenvalue.

Proof. By §10.5 take $T = L_A$ on $(\mathbb{R}^n, \text{dot product})$, $A = A^t$. On $\mathbb{C}^n$ with the Hermitian dot product take $AX = \lambda X$, $X \neq 0$, $\lambda \in \mathbb{C}$. As $A^* = A$, Observation 10.7.1 gives $\lambda\langle X, X\rangle = \langle AX, X\rangle = \langle X, A^*X\rangle = \langle X, \lambda X\rangle = \bar{\lambda}\langle X, X\rangle$, so $\lambda = \bar{\lambda} \in \mathbb{R}$. ■

Observation 15.3.3

If V has an orthonormal basis of eigenvectors of T, then T is symmetric.

Proof. $[T^t]_{\mathfrak{B}}^{\mathfrak{B}} = ([T]_{\mathfrak{B}}^{\mathfrak{B}})^t = [T]_{\mathfrak{B}}^{\mathfrak{B}}$ (Proposition 13.9.4, $[B]_{\mathfrak{B}} = I$, diagonal). ■

Theorem 15.3.4 (Spectral theorem, positive definite real case)

For $T \in \mathfrak{LM}$ (Notation 15.3.1) the following are equivalent: (1) T is symmetric; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1): Observation 15.3.3. (1)⇒(2): as Theorem 15.2.6 without conjugations; Proposition 15.3.2 gives a real unit eigenvector v; $W = \langle v\rangle^\perp$ is T-invariant ($\langle v, Tu\rangle = \langle Tv, u\rangle = \lambda\langle v, u\rangle = 0$); $T|_W$ is symmetric; induct. ■

Corollary 15.3.5 (Spectral theorem for real symmetric matrices)

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ (dot product on $\mathbb{R}^n$) the following are equivalent: (1) A is symmetric; (2) $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A; (3) $U^{-1}AU$ is diagonal for some $U \in \mathbf{O}(n)$.

Proof. Theorem 15.3.4 for L_A, with U the orthogonal matrix whose columns are the orthonormal eigenbasis. ■

Diagonalizing quadratic forms. With symmetric $J = (a_{ij})$, $2Q(X) = X^tJX$. Take $U \in \mathbf{O}(n)$ with $U^{-1}JU = D = \text{diag}(\lambda_i)$; as $U^t = U^{-1}$, $X = UX'$ gives

$$2Q(X) = X^tUDU^{-1}X = (U^{-1}X)^tD(U^{-1}X) = \lambda_1(x_1')^2 + \cdots + \lambda_n(x_n')^2 :$$

the quadratic form is **diagonalized**.

Example 15.3.6

For $J = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $2Q(x,y) = 2x^2 + 2xy + 2y^2$, and $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) J \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right)$. Substituting $x = \frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$, $y = -\frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$ gives $2x^2 + 2xy + 2y^2 = (x')^2 + 3(y')^2$.

Example 15.3.7

In the eigenbasis $\{v, w\} = \{\frac{1}{\sqrt{2}}(1,-1)^t, \frac{1}{\sqrt{2}}(1,1)^t\}$, $X = x'v + y'w$, the curve $2x^2+2xy+2y^2 = 4$ is $(x')^2 + 3(y')^2 = 4$: an ellipse rotated clockwise by $45°$.

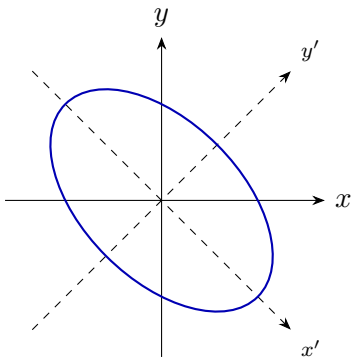


Figure. $2x^2 + 2xy + 2y^2 = 4$ *is the ellipse* $(x')^2 + 3(y')^2 = 4$ *in the rotated frame.*

Exercise 15.3.8

$x^2 + xy + y^2 - x - y + \frac{1}{3} \geq 0$ for all real x, y.

Solution. $x = u + \frac{1}{3}$, $y = v + \frac{1}{3}$ gives $u^2 + uv + v^2$, with matrix eigenvalues $\frac{3}{2}, \frac{1}{2} > 0$: $= \frac{3}{2}(u')^2 + \frac{1}{2}(v')^2 \geq 0$. □

Exercise 15.3.9

Sketch (a) $2x^2 - 2\sqrt{2}xy + 3y^2 = 4$;
(b) $x^2 - 6xy + y^2 = 2$;
(c) $11x^2 + 24xy + y^2 = 15$.

Solution. (a) $J = \left(\begin{smallmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 3 \end{smallmatrix}\right)$: eigenvalues $1, 4$, eigenvectors $(\sqrt{2}, 1)$, $(-1, \sqrt{2})$; ellipse $(x')^2 + 4(y')^2 = 4$. (b) $J = \left(\begin{smallmatrix} 1 & -3 \\ -3 & 1 \end{smallmatrix}\right)$: -2 at $(1,1)$, 4 at $(-1,1)$; hyperbola $4(y')^2 - 2(x')^2 = 2$. (c) $J = \left(\begin{smallmatrix} 11 & 12 \\ 12 & 1 \end{smallmatrix}\right)$: 19 at $(3,2)$, -7 at $(-2,3)$; hyperbola $19(x')^2 - 7(y')^2 = 15$. □

Exercise 15.3.10

Why are the high-school conics (with centre at the origin) exactly ellipses (including circles) and hyperbolas, plus parabolas?

Solution. Rotating axes (Corollary 15.3.5), $X^t JX + \ell(X) + c = 0$ has quadratic part $\lambda_1(x')^2 + \lambda_2(y')^2$; completing squares gives an ellipse if $\lambda_1\lambda_2 > 0$, a hyperbola if $\lambda_1\lambda_2 < 0$, a parabola if one $\lambda_i = 0$ (or degenerate cases). □

Exercise 15.3.11

For a finite-dimensional real quadratic space (V, B) with basis $\mathfrak{B}$, the following are equivalent: (1) B is positive definite; (2) all eigenvalues of $J = [B]_{\mathfrak{B}}$ are positive.

Solution. With $X = [v]_{\mathfrak{B}}$ and $J = UDU^t$, $U \in \mathbf{O}(n)$ (Corollary 15.3.5): $B(v,v) = \sum \lambda_i (x_i')^2$, $X' = U^t X$ arbitrary, positive for $X' \neq 0$ iff all $\lambda_i > 0$. (Answers Question 13.1.14.) □

Exercise 15.3.12

For a finite-dimensional Hermitian space (V, H) with basis $\mathfrak{B}$, the following are equivalent: (1) H is positive definite; (2) all eigenvalues of $J = [H]_{\mathfrak{B}}$ are positive reals.

Solution. $H(v,v) = X^t J \bar{X} = \overline{\bar{X}^t \bar{J} X}$, $X = [v]_{\mathfrak{B}}$. $\bar{J}$ is self-adjoint, so unitarily diagonalizable (Theorem 15.2.6) with real eigenvalues (as in Proposition 15.3.2), the same as those of J. So $H(v,v) = \sum \lambda_i |x_i'|^2$, positive for $X \neq 0$ iff all $\lambda_i > 0$. □

15.4 Orthogonal operators

Motivation. A real orthogonal operator splits V into orthogonal invariant lines and planes.

Here $F = \mathbb{R}$.

Proposition 15.4.1

Every $T \in \mathfrak{LM}$ over $\mathbb{R}$ has an invariant subspace W with $\dim W = 1$ or 2.

Proof. Let $m_T = p_1 \cdots p_s$ (monic irreducibles in $\mathbb{R}[t]$, of degree 1 or 2). Some $0 \neq v$ has $p_1(T)v = 0$: any $v \neq 0$ if $s = 1$, else $v = p_2(T) \cdots p_s(T) w \neq 0$ by minimality. If $p_1 = t + c$, $Tv = -cv$; if $p_1 = t^2 + at + b$, $T^2 v = -aTv - bv$. So $W = \langle v, Tv \rangle$ is T-invariant. ■

Theorem 15.4.2 (Spectral theorem for real orthogonal operators)

If V is a finite-dimensional real inner product space and T is orthogonal, there are subspaces $W_1, \ldots, W_s$ of V with (1) $V = W_1 \oplus \cdots \oplus W_s$; (2) each W_i is T-invariant; (3) $W_i \perp W_j$ for $i \neq j$; (4) $\dim W_i \in \{1, 2\}$.

Proof. Induction on $\dim V$. Take W as in Proposition 15.4.1; $V = W \oplus W^{\perp}$. T is injective, so $TW = W$, and for $w' = Tw \in W$, $v \in W^{\perp}$: $\langle w', Tv \rangle = \langle w, T^t T v \rangle = \langle w, v \rangle = 0$. So $T|_{W^{\perp}}$ is orthogonal; induct. ■

Corollary 15.4.3 (Spectral theorem for $\mathbf{O}(n)$)

For $A \in \mathbf{O}(n)$ there are $A_i \in \mathbf{SO}(2)$ and $U \in \mathbf{O}(n)$ with
$A \in \mathbf{SO}(n)$, $n = 2m$: $U^{-1} A U = \operatorname{diag}(A_1, \ldots, A_m)$;
$A \in \mathbf{SO}(n)$, $n = 2m+1$: $U^{-1} A U = \operatorname{diag}(1, A_1, \ldots, A_m)$;
$A \notin \mathbf{SO}(n)$, $n = 2m$: $U^{-1} A U = \operatorname{diag}(-1, 1, A_1, \ldots, A_{m-1})$;
$A \notin \mathbf{SO}(n)$, $n = 2m+1$: $U^{-1} A U = \operatorname{diag}(-1, A_1, \ldots, A_m)$.

Proof. (i) Theorem 15.4.2 for L_A gives an orthonormal basis $\mathfrak{B}$ from the W_i, $[I]_{\mathcal{E}}^{\mathfrak{B}} \in \mathbf{O}(n)$ (Observation 9.3.6(5)), and $[I]_{\mathcal{E}}^{\mathfrak{B}} \, A \, [I]_{\mathfrak{B}}^{\mathcal{E}} = [L_A]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(B_1, \ldots, B_s) \in \mathbf{O}(n)$ with B_i of size 1 or 2.

(ii) $\operatorname{diag}(B_1,\dots,B_s)\in\mathbf{O}(n)$ iff each $B_i\in\mathbf{O}(1)$ or $\mathbf{O}(2)$. So (a) $B_i=\pm1$ if $\dim W_i=1$;

(b) a reflection $B_i\in\mathbf{O}(2)-\mathbf{SO}(2)$ has $U_i^{-1}B_iU_i=\operatorname{diag}(-1,1)$, $U_i\in\mathbf{O}(2)$ (§9.4), so we may take $B_i=\operatorname{diag}(-1,1)$.

(iii) Reordering (a permutation matrix in $\mathbf{O}(n)$) gives $\operatorname{diag}(\pm1,\dots,\pm1,C_1,\dots,C_r)$, $C_j\in\mathbf{SO}(2)$. Pairs of 1's or -1's merge into $R_0=I_2$ or $R_\pi=-I_2$, leaving at most one 1 and one -1; as $\det A=(-1)^{\#(-1)}$, reordering gives the four forms. ■

Definition 15.4.4 (Maximal torus of SO(n))

(a) For $n=2m$: $\mathbf{T}_{\mathbf{SO}(n)}=\{\operatorname{diag}(A_1,\dots,A_m):A_i\in\mathbf{SO}(2)\}$. (b) For $n=2m+1$: $\mathbf{T}_{\mathbf{SO}(n)}=\{\operatorname{diag}(1,A_1,\dots,A_m):A_i\in\mathbf{SO}(2)\}$.

Exercise 15.4.5

(a) $\mathbf{T}_{\mathbf{SO}(n)}\le\mathbf{SO}(n)$.
(b) $\mathbf{T}_{\mathbf{SO}(n)}\approx\mathbf{SO}(2)\times\cdots\times\mathbf{SO}(2)\approx S^1\times\cdots\times S^1$ ($m=\lfloor n/2\rfloor$ factors).

Solution. Blockwise products and inverses, $\det=\prod\det A_i=1$; $\operatorname{diag}(\dots,A_i,\dots)\mapsto(A_i)$, and $\mathbf{SO}(2)\approx S^1$ (Example 11.9.19). □

Corollary 15.4.6 (Spectral theorem for SO(n))

If $A\in\mathbf{SO}(n)$, then $U^{-1}AU\in\mathbf{T}_{\mathbf{SO}(n)}$ for some $U\in\mathbf{SO}(n)$. Hence $\mathbf{SO}(n)=\bigcup_{U\in\mathbf{SO}(n)}U\,\mathbf{T}_{\mathbf{SO}(n)}\,U^{-1}$.

Proof. Corollary 15.4.3 gives $U\in\mathbf{O}(n)$ with $U^{-1}AU=\operatorname{diag}(1,R_{\theta_1},\dots,R_{\theta_m})$ (no leading 1 for even n; $n\ge2$). If $\det U=-1$, let $S=\operatorname{diag}(1,I_2,\dots,I_2,\operatorname{diag}(-1,1))$; since $\operatorname{diag}(-1,1)R_\theta\operatorname{diag}(-1,1)=R_{-\theta}$ (Observation 9.5.9), $(US)^{-1}A(US)=\operatorname{diag}(1,R_{\theta_1},\dots,R_{\theta_{m-1}},R_{-\theta_m})\in\mathbf{T}_{\mathbf{SO}(n)}$, and $US\in\mathbf{SO}(n)$. ■

Exercise 15.4.7

In Corollary 15.4.3, can U be chosen in $\mathbf{SO}(n)$ even when $A\notin\mathbf{SO}(n)$?

Solution. Yes: replacing the basis vector u of the block -1 by $-u$ flips $\det U$ and leaves $U^{-1}AU$ unchanged. □

So in a suitable orthonormal frame an orthogonal operator rotates planes and flips at most one axis; Examples 9.6.7 and 9.6.10 are the typical cases.

Exercise 15.4.8

For $A\in\mathbf{SO}(n)$ below, find $U\in\mathbf{SO}(n)$ with $U^{-1}AU\in\mathbf{T}_{\mathbf{SO}(n)}$: (a) $A=\left(\begin{smallmatrix}0&1&0\\1&0&0\\0&0&-1\end{smallmatrix}\right)$;
(b) $A=\left(\begin{smallmatrix}0&1&0&0\\1&0&0&0\\0&0&0&-1\\0&0&-1&0\end{smallmatrix}\right)$.

Solution. (a) $Au_1=u_1$ for $u_1=\frac{1}{\sqrt2}(1,1,0)^t$, and $A=-I$ on $\langle u_2,u_3\rangle$, $u_2=\frac{1}{\sqrt2}(-1,1,0)^t$, $u_3=e_3$. $U=(u_1,u_2,u_3)$ has $\det U=1$ and $U^{-1}AU=\operatorname{diag}(1,R_\pi)$. (b) Eigenvalue 1: $u_1=\frac{1}{\sqrt2}(1,1,0,0)^t$, $u_2=\frac{1}{\sqrt2}(0,0,1,-1)^t$; eigenvalue -1: $u_3=\frac{1}{\sqrt2}(1,-1,0,0)^t$, $u_4=\frac{1}{\sqrt2}(0,0,1,1)^t$. With $U=(u_1,u_2,u_3,u_4)$, $U^{-1}AU=\operatorname{diag}(R_0,R_\pi)$; $\det U=1$ (if it were -1, replace u_4 by $-u_4$, which keeps $R_\pi=-I_2$). □

15.5 Epilogue

Motivation. Without positive definiteness the spectral theorems fail; Schur and the perp theorem survive.

§§15.2–15.4 needed Gram–Schmidt: if $Tv = \lambda v$ with $B(v,v) = 0$, $V = \langle v\rangle \oplus \langle v\rangle^\perp$ fails.

A false proof. "For normal T on non-degenerate (V,H), induct via $\bar{V} = V/\langle v\rangle$ with $\bar{H}(\bar{u},\bar{w}) = H(u,w)$." But $\bar{H}$ is well defined only if $v \perp V$. The statement is false:

Example 15.5.1

For $J = \operatorname{diag}(1,-1)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$ in $(\mathbb{R}^2, B^J_{\mathcal{E}})$, L_A is symmetric (Exercise 13.9.6(b)), but $\phi_A(t) = (t-2)^2$ and $A \neq 2I$, so L_A is not diagonalizable.

Example 15.5.2

For $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ in $(\mathbb{C}^2, H^J_{\mathcal{E}})$: $H^J_{\mathcal{E}}(A(x,y)^t,(z,w)^t) = (x+y)\bar{w} + y\bar{z} = H^J_{\mathcal{E}}((x,y)^t, A(z,w)^t)$, so L_A is self-adjoint, yet not diagonalizable.

Theorem 15.5.3 (Schur)

If $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional Hermitian inner product space, then for every $T \in \mathfrak{LM}$ there is an orthonormal basis $\mathfrak{B}$ with $[T]^{\mathfrak{B}}_{\mathfrak{B}}$ upper triangular.

Proof. Take a unit eigenvector $T^*v = \mu v$ (for $T = L_A$, Observation 10.7.1 gives T^*). $\langle v\rangle^\perp$ is T-invariant: $\langle v, Tw\rangle = \langle T^*v, w\rangle = \mu\langle v,w\rangle = 0$. By induction it has an orthonormal basis $\mathfrak{C}$ triangularizing T; put $\mathfrak{B} = \mathfrak{C} \cup \{v\}$, v last. ■

(The proofs of §7.3, §12.5 work over any algebraically closed field.)

Coding theory. $\mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$ is a field ($\bar{1}\cdot\bar{1} = \bar{1}$).

Definition 15.5.5 (Binary codes)

A $\mathbb{Z}_2$-subspace C of $(\mathbb{Z}_2)^n$ is a **linear binary code of length** n; with the ordinary inner product $B^I_{\mathcal{E}}$ on $(\mathbb{Z}_2)^n$, $C^\perp$ is the **dual code**.

Theorem 15.5.6

If C is a linear code in $(\mathbb{Z}_2)^n$, then $\dim C + \dim C^\perp = n$; so if $\dim C = m$, $|C| = 2^m$ and $|C^\perp| = 2^{n-m}$.

Proof. $B^I_{\mathcal{E}}$ is non-degenerate ($\det I = 1$): the perp theorem (Theorem 13.5.11 or 13.8.11) holds over any field, though Gram–Schmidt fails ($B^I_{\mathcal{E}}((\bar{1},\bar{1})^t,(\bar{1},\bar{1})^t) = \bar{0}$). ■